

THERMAL BOARD INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY.

- A. Work described in this section continuous exterior thermal board insulation for Rainscreen wall application.
- B. Related work specified elsewhere:
 - 1. Section 074213 – Metal Plate Wall Panels
 - 2. Section 072650 – Fluid Applied Air and Water Barrier

1.3 DEFINITIONS

- A. American Society for Testing and Materials (ASTM):
 - 1. C165-07: Standard Test Method for Measuring Compressive Properties of Thermal Insulations.
 - 2. C177-10: Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of Guarded-Hot-Plate Apparatus.
 - 3. C209-07ae1: Standard Test Methods for Cellulosic Fiber Insulating Boards.
 - 4. C356-10: Standard Test Method for Linear Shrinkage of Preformed High-Temperature Thermal Insulation Subjected to Soaking Heat.
 - 5. C411-11: Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation.
 - 6. C423-09a: Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.
 - 7. C518-10: Standard test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
 - 8. C612-10: Standard Specification for Mineral Fiber Block and Board Thermal Insulation.
 - 9. C665-12: Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing.
 - 10. C795-08: Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel.
 - 11. C1104-00(2006): Standard Test Method for Determining the Water Vapor Sorption of Unfaced Mineral Fiber Insulation.
 - 12. D2126-09: Standard Test Method for Response of Rigid Cellular Plastics to Thermal and Humid Aging.

13. E84-12: Standard Test Method for Surface Burning Characteristics of Building Materials.
14. E96-10: Standard test Method for Water Vapor Transmission of Materials.
15. E136-11: Standard Test Method for Behavior of Materials in a Vertical Tube Furnace at 750°C.

B. National Fire Protection Association (NFPA):

1. NFPA 268-2012: Standard Test Method for Determining Ignitability of Exterior Wall Assemblies Using a Radiant Heat Energy Source.
2. NFPA 285-2012: Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Non-Load Bearing Wall Assemblies Containing Combustible Components.

C. Underwriters Laboratory of Canada (ULC):

1. ULC S114-05: Standard Test Method for Determination of Non-Combustibility in Building Materials.

1.4 DESIGN AND PERFORMANCE CRITERIA.

A. General Performance: Continuous exterior rigid mineral fiber board thermal insulation shall satisfy all requirements of ASTM E612, Type IV-B, shall be free of defects, and meet each of the performance requirements specified herein.

B. Thermal Resistance:

1. At 25°F (-4°C), the R-value (RSI-value) per inch (25.4mm) thickness shall be 4.3 hr·ft²·°F/BTU (0.74 m²K/W) minimum, in accordance with ASTM C518/C177.
2. At 75°F (24°C), the R-value (RSI-value) per inch (25.4mm) thickness shall be 3.9 hr·ft²·°F/BTU (0.68 m²K/W) minimum, in accordance with ASTM C518/C177.

C. Fire and Heat Performance:

1. Surface Burning Characteristics: Flame spread rating shall be 0 and smoke development rating shall be 0 when tested in accordance with ASTM E84.
2. Combustibility: The board insulation shall be rated as “Non-Combustible” when tested in accordance with ULC S114.
3. The board insulation shall not ignite, burn, support combustion, or release flammable vapors when subjected to fire or heat at 1,382°F (750°C), in accordance ASTM E136.
4. The board insulation shall be considered Non-Combustible when tested in accordance with NFPA 268, and shall be deemed to be acceptable for use in exterior walls of Type I, II, III, and IV building construction without any constraints, such as use of a thermal barrier, as may be required by NFPA 285.
5. The board insulation shall be rated for a 1,200°F (650°C) service temperature, as determined by testing in accordance with ASTM C411.

D. Moisture Resistance:

1. Water absorption, as measured in accordance with ASTM C209, shall not exceed 1.2%.

2. Water vapor sorption, as measured in accordance with ASTM C1104, shall not exceed 0.30%.
 3. Water vapor transmission, as measured in accordance with ASTM E96, shall be at least 40 perms (2,280 SI-perms).
- E. Dimensional Stability and Physical Properties:
1. The board insulation shall have a minimum density of 11.0 pcf (176 kg/m³) as measured in accordance with ASTM C612.
 2. The board insulation shall have a minimum compressive strength of 1,220 psf (58.5 kPa) at 10% maximum deformation when tested in accordance with ASTM C165.
 3. The board insulation shall have a maximum of 0.38% linear shrinkage at 1,200°F (650°C) when tested in accordance with ASTM C356.
 4. The board insulation shall exhibit dimensional stability of a maximum linear change of 0.1% after 7 days in accordance with ASTM D2126 at each of the following climate conditions:
 - a. 40°F (-4°C) with ambient relative humidity;
 - b. 200°F (93°C) with ambient relative humidity;
 - c. 158°F (70°C) with 97% relative humidity.
- F. Acoustical Performance: When tested in accordance with ASTM C423, a 2-inch (51mm) insulation board shall have a NRC rating of 0.85 minimum.
- G. Corrosion Resistance:
1. The insulation board shall be rated "Non-corrosive" to steel when tested in accordance with ASTM C665.
 2. The insulation board shall conform with the requirements of ASTM C795 with respect to corrosion resistance in contact with stainless steel materials.

1.5 SUBMITTALS.

- A. General, Rainscreen Wall Assembly Components: Complete submittals shall be made jointly and simultaneously for all components of the Rainscreen wall assembly, including:
1. Exterior wall sheathing board, if applicable;
 2. Air and water resistive barrier;
 3. Rainscreen wall continuous exterior insulation;
 4. Metal rainscreen wall cladding panels and subframing components;
 5. All other trim, flashing, sealants, and components necessary for a complete rainscreen wall assembly as required by these specifications.
- B. Shop drawings.

1. Show complete rain screen wall system with air and water barrier(s), vapor retarder (if applicable), continuous exterior insulation, subframing system, metal cladding panels, ventilation components, flashings and accessories in elevation, sections, and details. Include metal thicknesses and finishes, panel lengths, joining details, anchorage details, flashings and special fabrication provisions for termination and penetrations. Indicate relationships with adjacent and interfacing work.
 2. All components shall be integrated into a single comprehensive and compete shop drawing set prepared by the metal cladding system manufacturer.
 3. Shop drawings shall identify each product and component by manufacturer, product name, and thickness, size, style, or other uniquely distinguishing characteristics.
- C. Warranty: Provide unexecuted specimen warranty documents for each warranty as required in specification article 1.10.
- D. Design Test Reports.
1. Submit copies of design test reports for each of the performance testing standards listed in specification article 1.4.
 2. Test reports shall be performed by independent, accredited testing laboratories, and shall bear the seal of a registered professional engineer.
- E. Samples.
1. Submit three (3) samples of rigid board insulation, minimum of 6" x 6" in size of the same thickness and properties, including facers, if any, as is specified for this project.
 2. Submit sample of insulation fasteners and all other hardware and system components as may be required for a complete installation.

1.6 QUALITY CRITERIA/INSTALLER QUALIFICATIONS.

- A. Engage an experienced wall system contractor (erector) to install insulation system who has a minimum of three (3) years of experience specializing in the installation of continuous exterior board insulation in a Rainscreen type application.
- B. Contractor must be certified by manufacturer specified as a supplier of the insulation system and obtain written certification from manufacturer that installer is approved for installation of the specified system.
- C. Successful contractor must obtain all components of Rainscreen wall system from a single manufacturer. Any secondary products that are required which cannot be supplied by the specified manufacturer must be recommended and approved in writing by primary manufacturer prior to bidding.
- D. Fabricator/Installer shall submit work experience and evidence of adequate financial responsibility. Architect reserves the right to inspect fabrication facilities in determining qualifications.

1.7 DELIVERY, STORAGE, AND HANDLING.

- A. Inspect materials upon delivery.
- B. Handle materials to prevent damage.

- C. Store materials off ground providing for drainage; under cover from weather; providing for air circulation; and, protected from any debris.

1.8 PROJECT CONDITIONS

- A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit insulation work to be performed according to manufacturer's written instructions and warranty requirements.
- B. Field Measurements: Verify actual dimensions of construction contiguous with by field measurements before fabrication/installation.

1.9 COORDINATION

- A. Coordinate sizes and locations of windows, doors, and wall penetrations with actual equipment provided.
- B. Coordinate insulation system with wall sheathing, masonry, air and water resistive barriers, metal wall cladding, rain drainage work, flashing, trim, and construction of other adjoining work to provide a leak proof, secure, and noncorrosive installation.

1.10 WARRANTIES

- A. Endorse and forward to owner the following warranties:
 - 1. Special Manufacturer's Rainscreen Wall Assembly Warranty: The mineral fiber rigid board insulation must be approved for use in the Rainscreen wall assembly in conjunction with the air and water resistive barrier and exterior metal cladding system; the use of specified mineral fiber rigid board insulation shall not nullify any manufacturers' warranties required elsewhere in this specification.
 - 2. Installer's 5 year warranty covering insulation system installation and workmanship.
- B. Warranties shall commence on date of substantial completion.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Mineral fiber rigid board insulation.
 - 1. All products are subject to full compliance with specification requirements, and shall be submitted for approval.
 - 2. Material: Basalt rock and slag mineral fiber board insulation, Type IV-B in accordance with ASTM C612.

3. Thickness: 1 layer of 2-inch thickness
4. Insulation board size shall be 48-inches (1,219mm) x 72-inches (1,219mm).

2.2 MISCELLANEOUS MATERIALS

- A. Insulation board attachment fasteners: Corrosion resistant steel screws, #10 minimum diameter x length appropriate for substrate. Use four (4) screws minimum per insulation board.

2.3 FABRICATION

- A. Use full size insulation boards to the greatest extent possible. As necessary size boards to fit at openings and elevation terminations with neat and straight cuts. Size boards to fit tightly together without gaps or voids.

PART 3 - PREPERATION & EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, sheathing and air/water resistive barriers, and other conditions affecting performance of the Work.
- B. Examine solid wall sheathing to verify that sheathing joints are supported by framing or blocking and that installation is within flatness tolerances required by insulation system and metal wall panel manufacturers.
- C. Examine roughing-in for components and systems penetrating wall to verify actual locations of penetrations relative to seam locations of insulation before beginning insulation installation.
- D. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.
- E. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Clean substrates of substances harmful to insulation, including removing projections capable of interfering with insulation attachment.
- B. Establish straight, side and crosswise benchmarks
- C. All walls shall be checked for square and straightness. Inside and outside corners may not be plumb; set a true line for the corner flashing with string line.
- D. Use proper size and length fastener for strength requirements to attach insulation boards. Attachment of insulations boards shall be done in a manner sufficient to adequately secure the insulations to the wall substrate until such time as the exterior cladding is fully secured in place.
- E. Installed system shall be true to line and plane and free of damage and physical defects.
- F. Remove damaged work and replace with new, undamaged components.

G. Leave Work in clean condition at date of substantial completion.

3.3 ERECTION TOLERANCES

A. Installation Tolerances: Install insulation board units within installed flatness tolerance of no more than 1/4 inch in 20 feet (6 mm in 6 m) at location lines as indicated and within nor more than 1/8-inch (3-mm) offset of adjoining edges. Gap between boards shall be no more than 1/8-inch (3-mm). Insulation boards shall be installed in continuous contact with back-up wall exterior surface.

3.4 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect insulation installation, including accessories. Report results in writing.
- B. Remove and replace applications of insulation boards where inspections indicate that they do not comply with specified requirements.
- C. Additional inspections, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

3.5 CLEANING

- A. Remove temporary protective coverings and packaging materials, if any, as insulation boards are installed unless otherwise indicated in manufacturer's written installation instructions. Maintain in a clean condition during construction.
- B. Replace insulation boards that have been damaged or have deteriorated beyond successful repair procedures.

END OF SECTION